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WHITE PAPER

PSEQ WHITE PAPER SERIES · PAPER 1 OF 4

PSeq: Whole-Molecule Phased RNA Sequencing on Standard Short-Read NGS

Platform architecture, workflow, and data model

July 2026 | Version 1.0

Executive summary

PSeq is a research-use only RNA-sequencing platform that brings whole-molecule resolution and end-to-end phasing to standard short-read next-generation sequencing (NGS). During reverse transcription, a source-molecule tagging reagent labels each RNA sequence, and a library strategy preserves the tag marker across random fragments. Paired-end sequencing runs on unmodified instruments, while an automated pipeline groups reads by Source Molecule Identifier (SMID), identifies the source gene, assembles a consensus sequence, and retains supporting records in a run-specific SQL database.

This paper, and the three papers that follow, presents PSeq's core innovation, architecture, and operating model.

> **Central proposition:** PSeq is designed to combine the throughput and base-level accuracy of short-read NGS with molecule-resolved, end-to-end transcript reconstruction and evidence traceability.

PSeq at a glance

- Runs on unmodified paired-end short-read NGS instruments.
- Assigns a high-diversity SMID and strand-sense discriminator (SD) before library fragmentation.
- Groups fragment reads by their source-molecule tag before gene identification and transcript assembly.
- Retains raw reads, intermediate analyses, alignments, consensus sequences, and run metadata in a sample-run database.

The short-read linkage problem

Conventional RNA-Seq achieves high throughput by sequencing short fragments prepared from a heterogeneous RNA population. It can identify expressed genes, quantify read abundance, and discover exon junctions. However, when two sequence features are separated by more than the sequenced fragment read length, the data no longer directly report whether those features occurred on the same source molecule.

Transcript identity and isoform abundance are therefore commonly inferred from populations of fragments using statistical models and sample ground-truth prior information. Those approaches are powerful, but the inferred transcript model can conflate evidence from multiple molecules and does not preserve an experimentally assigned identity for each original RNA.

Long-read instruments address linkage more directly but differ from short-read NGS in throughput, operating model, and error characteristics. PSeq approaches the linkage problem from the opposite direction: retain the accuracy of long-read and throughput of short-read sequencing but add a physical source-molecule identity that survives library preparation and guide computational reconstruction with better accuracy and scale.

PSeq Clear-Signal Architecture™

PSeq Clear-Signal Architecture™ is the integrated molecular and computational design used to preserve source-molecule identity from reverse transcription through sequence reconstruction. A Source Molecule Identifier and strand-sense information are introduced before fragmentation, retained across fragments derived from the same tagged cDNA, and recovered after sequencing. The data pipeline then groups reads by molecular identity before gene identification and transcript assembly, while retaining links to the underlying read, alignment, and quality evidence.

The architecture is based on three linked principles:

- **Identity is established before fragmentation.**
- **Reads are grouped by source molecule before reconstruction.**
- **The reconstructed molecule remains connected to its supporting evidence.**

Together, these principles are intended to convert a mixed short-read data set into distinct, independently inspectable molecular records.

PSeq design innovation principles

- Identify before fragmenting. A source-molecule tag is introduced during reverse transcription, before the cDNA is converted into a sequencing library.
- Preserve molecular provenance. Random fragments derived from the same tagged cDNA carry the same marker block and can be grouped after sequencing.
- Separate identity from sequence inference. The SMID establishes the molecular grouping; gene identification and data assembly then operate within that group.
- Retain the evidence chain. Final molecular records are designed to remain traceable through alignments and intermediate files to the originating FASTQ records and base-quality information.
- Scale on familiar infrastructure. Library sequencing uses paired-end NGS, while computation is designed to scale through multithreaded cloud resources.

End-to-end operating model

At the laboratory level, heterogeneous RNA is reverse transcribed in the presence of a multifunctional tagging reagent. Each tag contains a high-diversity SMID and structural elements that report strand sense. The tagged reverse transcript is amplified, fragmented through reactions intended to remain intramolecular, coupled to conventional sequencing adaptors, and converted into a paired-end PSeq DNA library.

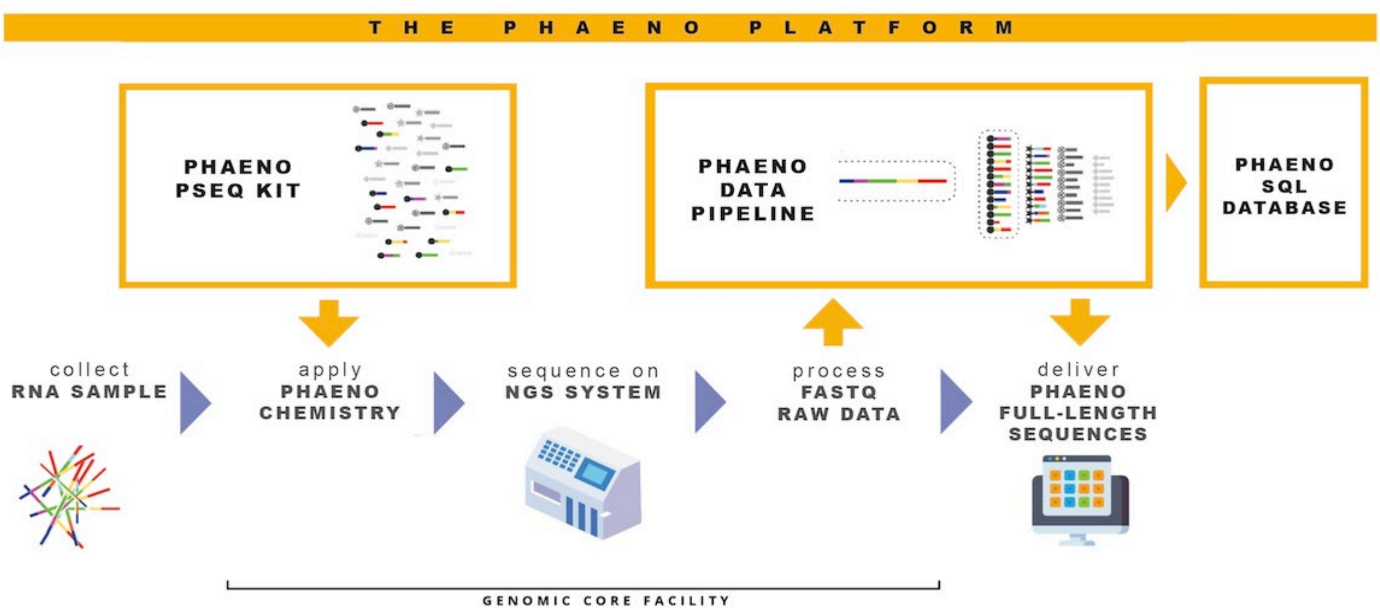

Figure 1. PSeq fits a familiar genome-core workflow: proprietary library preparation and automated data assembly are added around conventional paired-end NGS.

After sequencing, raw FASTQ files enter the PSeq data pipeline. Marker blocks are localized, reads are clustered by SMID, tag sequence is trimmed, strand sense is recorded, redundant reads are removed, and the remaining sequences are used for source-gene identification and transcript assembly. Intermediate and final records populate a run-specific SQL database.

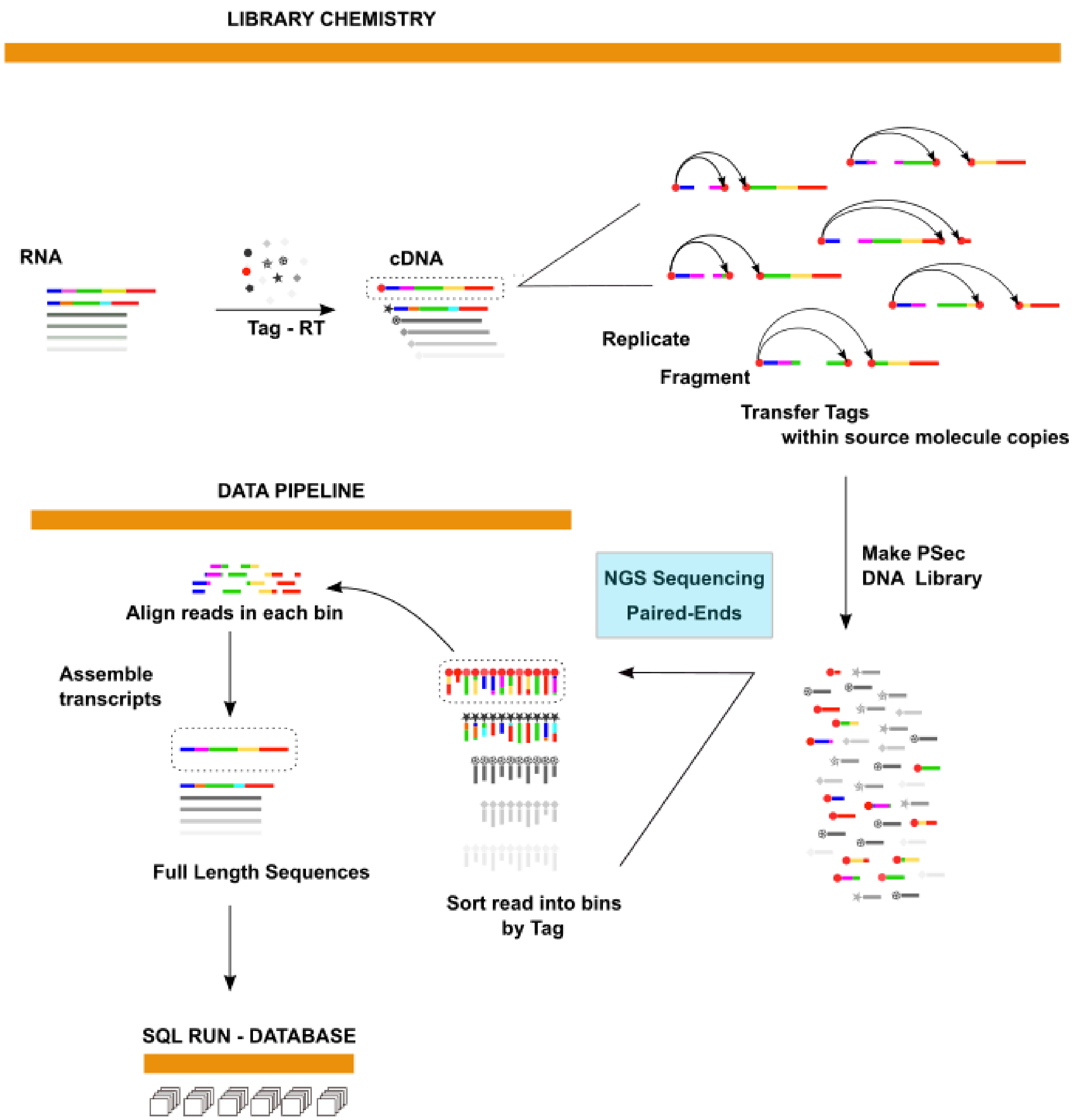

Figure 2. Conceptual PSeq workflow from RNA tagging and library construction through paired-end sequencing, SMID-based read grouping, sequence assembly, and SQL run-database output.

Three coupled platform components

Molecular tagging and library chemistry

The wet-lab component is designed to mark each reverse-transcribed source RNA with a SMID and a strand-sense discriminator (SD). During later fragmentation, a marker from the tagging reagent is redistributed to random fragments derived from that same cDNA. The resulting short-read library will therefore both local sequence information and orginal molecular identity.

Automated data pipeline

The computational pipeline converts a mixed population of read pairs into molecule-specific bins. Within each SMID bin, the data pipeline identifies the most likely source gene, retrieves relevant reference gene sequence, performs de novo and reference-guided assembly, and generates alignment and consensus products.

Run-specific SQL data product

The SQL database is not simply an index of final sequences. It is intended to document the run, the source sample, the molecule-level groupings, the intermediate transformations, and the supporting evidence for every reconstructed RNA. The sample record set includes client and run metadata, FASTQ references, SMID and strand information, gene identity, assembly products, FASTA and BAM output, consensus sequences, and links to external gene resources.

What changes when the molecule becomes the unit of analysis

A conventional expression profile aggregates evidence across a population. PSeq is designed to add a second level of analysis in which each source molecule has its own evidence bundle. The new level the questions the data system can support:

- Which reads contributed to this reconstructed molecule?
- Which gene and strand were assigned to the molecule, and on what evidence?
- Which splice junctions occur together from end to end?
- Where do individual reads disagree with the consensus or reference sequence?
- Can an unusual base call, junction, or assembly decision be traced back to its original FASTQ record and quality score?

Deployment model

PSeq operates within conventional sequencing laboratories and genome cores. The platform does not require modification of the sequencer; the differentiated elements are the library reagents and protocols, the automated data pipeline, and the run-specific database. The initial implementation described for PSeq v1 uses Amazon EC2 resources, local reference databases, and multithreaded processing. The architecture is designed to scale vertically through additional cores and horizontally through multiple instances.

A 6 to12-hour analysis cycle is expected, however actual processing time will depend on run size, compute configuration, reference resources, and the depth of per-molecule assembly and quality control.

PSeq Deliverables

Client meta data (FASTA)

Run meta data (FASTA)

Demultiplexed raw files (FASTQ)

Tagged FASTQ records

Transcript inventory (.gtf)

- Novel isoforms
- Registered isoforms

Source Gene Inventory (.gtf)

- Novel
- Registered, expanded annotation
- Registered, unmodified

Isoform sequences (.bam)

- Consensus
- trimmed reads

RSeQC (.html)

Deliverables summarized in a PDF report.

Intended context and current boundaries

PSeq v1 is initially being configured for research-use only (RUO) implementation. Detailed kit compositions, reagent specifications, laboratory recipes, and operational quality controls will be archived in design control documentation. Likewise, performance data will be presented separately.

The PSeq workflow is currently configured to bulk RNA, whole-genome sequencing, whole-exome sequencing, and RNA-Seq laboratory environments. We believe PSeq may be adaptable to single-cell workflows, this hypothesis will be tested soon.

Interpretation boundary: PSeq should be evaluated as an integrated chemistry-and-computation system. Platform diagrams describe the intended workflow; they do not by themselves establish accuracy, reproducibility, sensitivity, or fitness for a clinical purpose.

Conclusion

PSeq reframes short-read RNA sequencing around the source molecule. A physical tag links random fragments to one reverse transcript; the data pipeline reconstructs and documents the molecule; and a run-specific database retains the supporting evidence. If the molecular labeling, read grouping, and reconstruction steps perform as intended, the result is a scalable route to phased, molecule-resolved RNA records using familiar sequencing infrastructure. The companion white papers in this series describe the molecular design, the computational implementation, and the initial evidence supporting that integrated chain.

Notes

1. Instrument examples in the source report include Illumina short-read systems, Complete Genomics DNB sequencers, and Element Biosciences AVITI instruments; the list is illustrative rather than exhaustive.
2. The platform description is derived from the internal PSeq v1 overview and technical evidence report dated July 21, 2026. Operational specifications remain subject to the controlled SOP set.
3. Research-use context only. This paper does not establish clinical validity or regulatory clearance.
4. RSeQC is a comprehensive open-source RNA-Seq Quality Control package that is modified for PSeq data structure. (see https://rseqc.sourceforge.net/index.html).

PSeq white paper series

PAPER 1	PAPER 2	PAPER 3	PAPER 4
PSeq: Whole-Molecule Phased RNA Sequencing on Standard Short-Read NGS	PSeq Molecular Tagging and Library Architecture	PSeq Data Pipeline	Initial Technical Validation of the PSeq Whole-Molecule RNA Sequencing Platform.